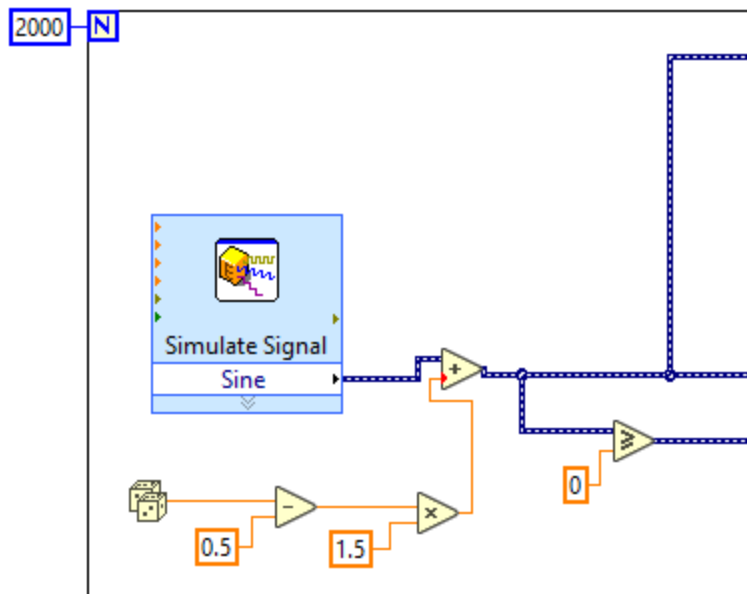
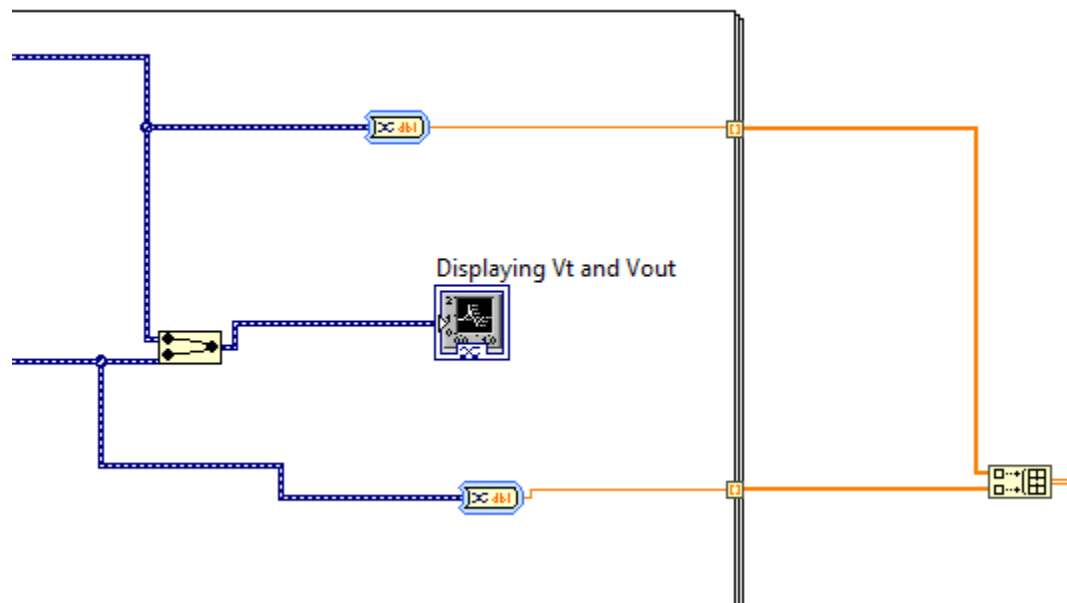


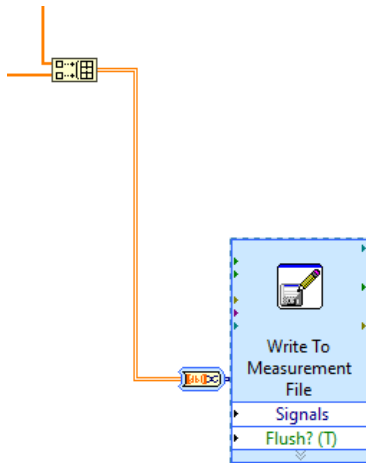
I replaced the while loop with a for loop, making sure to delete the stop button and put 2000 for the number of iterations.



I connected the outputs V_{out} and V_T to an array outside of the loop, making sure that tunnel mode was set to indexing rather than last value.



I used the “write MEAS file” and I connected it to the array to save the data to an Excel file. I also employed a couple FROM DDT and a TO DDT to get the individual numbers to the array proper, then adjusting the array to input in the “write MEAS file” correctly. Once saved to an external Excel file the data was easy to evaluate to verify the values were correct.



Configure Write To Measurement File [Write To Measurement File]

Filename
 C:\Users\clwilson47\Documents\MATLAB\Lab5_Data.xlsx

Action
☒ Save to one file
☐ Ask user to choose file
 ☒ Ask only once
 ☐ Ask each iteration
If a file already exists
☒ Rename existing file
☐ Use next available filename
☐ Append to file
☐ Overwrite file
☐ Save to series of files (multiple files)
 Settings...

File Format
☐ Text (LVM)
☐ Binary (TDMS)
☐ Binary with XML Header (TDM)
☒ Microsoft Excel (.xlsx)
☒ Lock file for faster access

Segment Headers
☒ One header per segment
☐ One header only
☐ No headers

X Value (Time) Columns
☐ One column per channel
☐ One column only
☒ Empty time column

Delimiter
☒ Tabulator
☐ Comma

File Description
 Advanced...

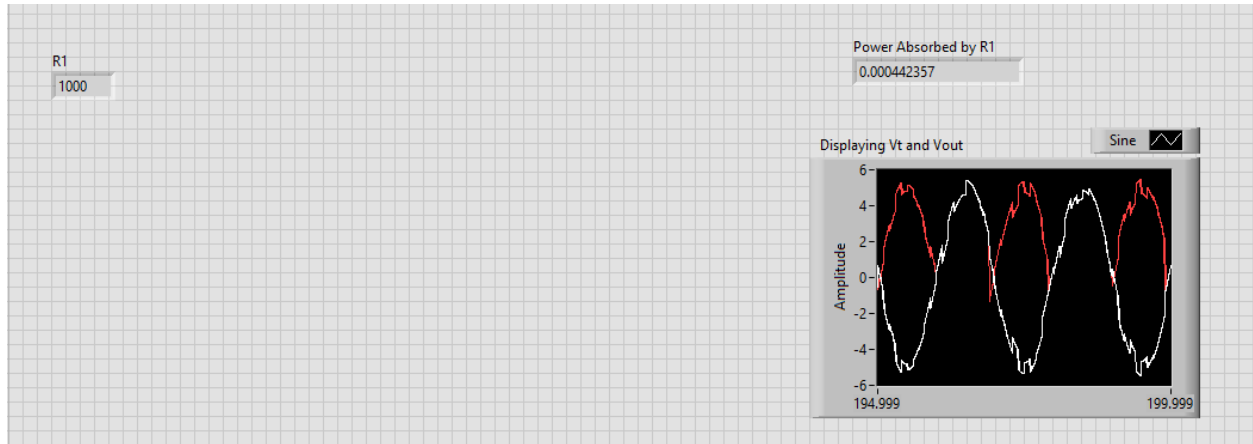
OK Cancel Help

The Vout is always positive as expected:

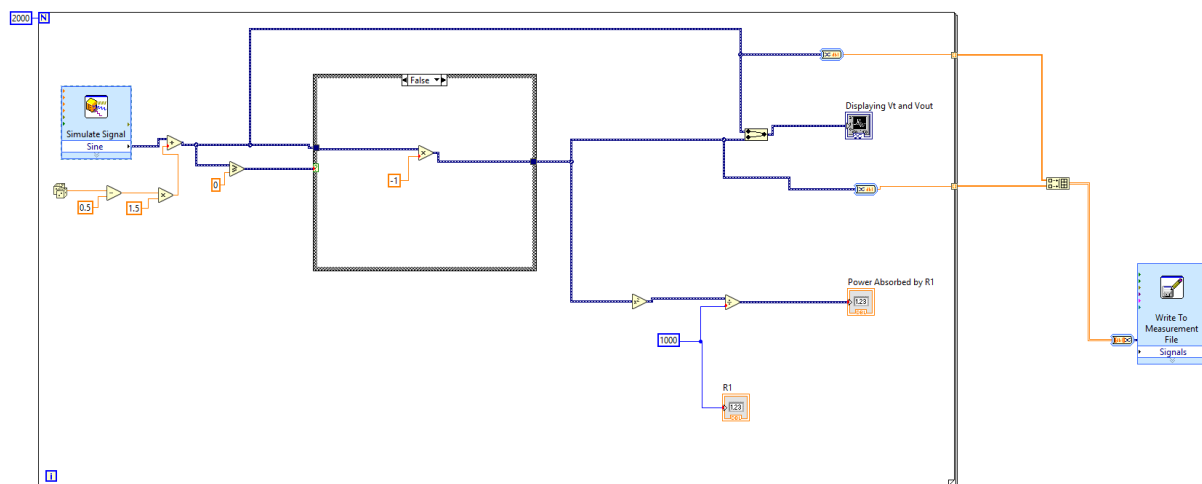
Vt	Vout
1.403392	1.403392
3.209813	3.209813
3.307951	3.307951
4.863907	4.863907
4.614841	4.614841
4.65583	4.65583
3.520766	3.520766
2.909708	2.909708
1.251025	1.251025
0.428811	0.428811
-2.17907	2.179072
-3.41483	3.41483
-3.33391	3.333912

My finalized program looked like this:

Front Panel:



Block Diagram:



Conclusion

This lab was interesting learning how to save data to files. It was simple to implement since most of the analysis of the circuit and such was done last lab, so we could focus on the newer stuff like saving to files and using arrays. This will be useful in the future for analyzing data using MATLAB, Excel, or other programs. This will probably become even more applicable whenever we start using the DAQs more in this laboratory and in real life applications to analyze signals.